

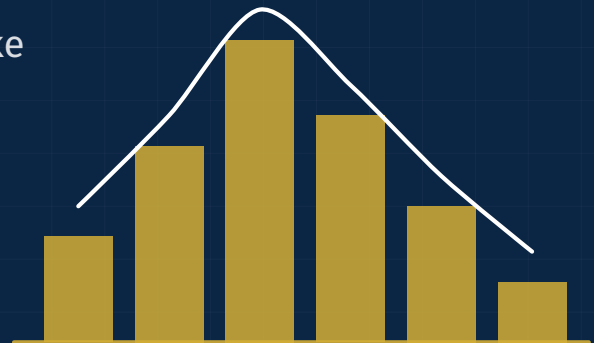


DR ESLAM AHMED
eliteigcse.com

CHAPTER 6 • STATISTICS & PROBABILITY • TOPIC 4

Probability Toolkit

All the probabilities add to 1. Expected number = probability $\times$ trials. For more than one pick, multiply along — and take one away from the bag if it is not replaced.



4

VARIANTS

32

PAST-PAPER PARTS

42

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Tables: total 1

V2 Expected frequency

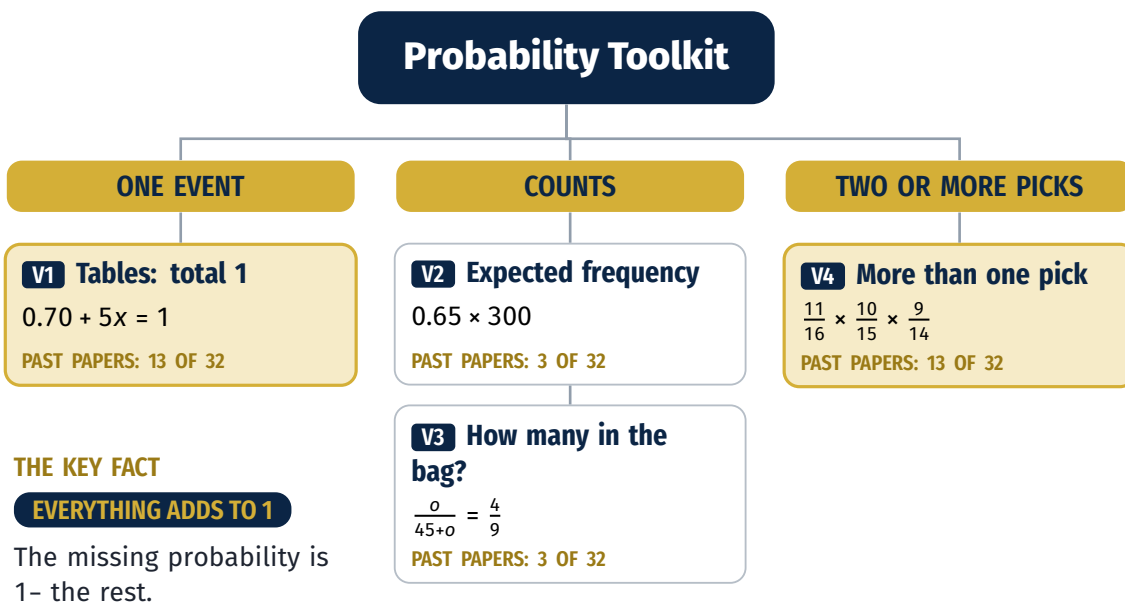
V3 How many in the bag?

V4 More than one pick

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Probability at a glance



Your notes


★ How often does each variant come up?

31 past-paper questions (Jan 2020 → Nov 2025) hold 32 parts; 42 practice parts — grouped by type. Tree and Venn diagrams are in Topic 5.

	PAST PAPERS (OUT OF 32)	PRACTICE PARTS
V1 Tables: total 1	13	14
V2 Expected frequency	3	13
V3 How many in the bag?	3	9
V4 More than one pick	13	6

WHAT THIS TELLS YOU Two kinds of question: an early probability table (Q1–Q6, 3–5 marks, “add to 1, then times the trials”) and a late multi-pick question (Q15–24, 3–6 marks, often with algebra).

BEFORE YOU START

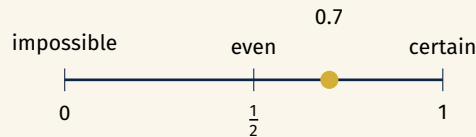
The big idea



Probability is a fraction of the whole.

$P(\text{event}) = \frac{\text{ways it can happen}}{\text{all equally likely ways}},$
between 0 and 1.

All the outcomes together: **total 1**. Not happening: $1 - P$.


The rules
Total

$\sum P = 1$
All outcomes.

Not

$1 - P(A)$
The complement.

OR

$P(A) + P(B)$
Can't both happen: add.

AND

$P(A) \times P(B)$
One after another: multiply.

Expected number
Expected frequency

$P \times \text{trials}$
An estimate, not a promise.

Relative frequency

$\frac{\text{times it happened}}{\text{trials}}$
Best estimate: the most trials.

**CALCULATOR FIRST, THEN CHECK**

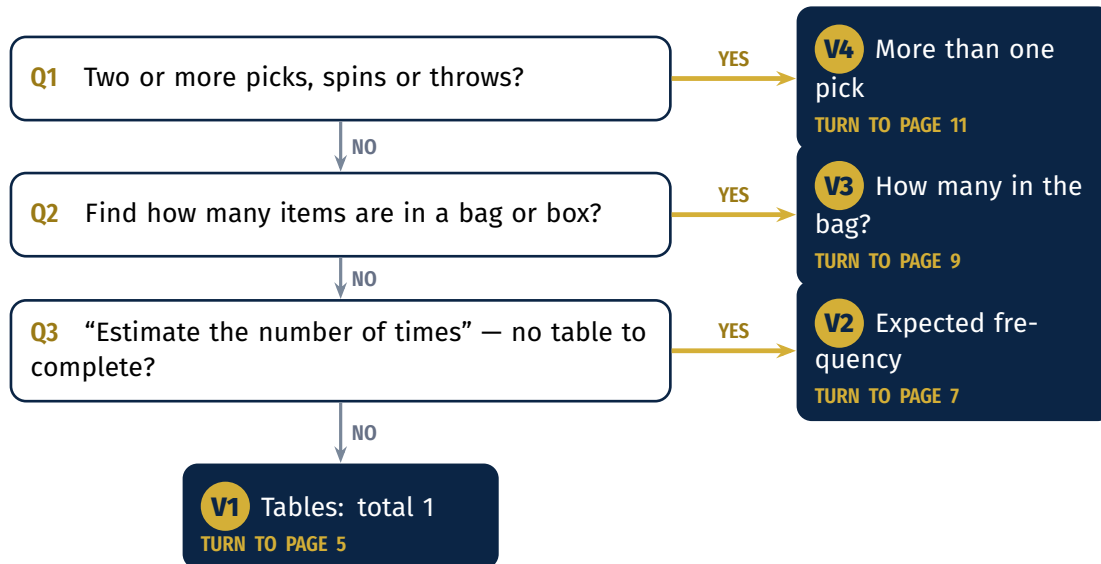
Fractions: use the fraction key and leave answers like $\frac{33}{112}$ unless decimals are asked.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | |
|---|---|
|  What is it? |  When you see → you do |
|  How do I do it? |  Watch out |
|  Worked example |  Exam tip |
|  Check yourself |  Calculator check |

V1

VARIANT 1 MOST COMMON IN THE EXAM

SEEN IN PAST PAPERS

13 / 32

PRACTICE QUESTIONS: 14

Probability Tables: Total 1

Missing values make the total 1 (use x if needed); then $\times$ the number of trials.

Came up in: Jan 2020 P2H Q6 • May 2021 P1H Q6 • Jan 2022 P2HR Q1 • Jan 2022 P2H Q6 • Jun 2022 P1HR Q1 • Jun 2022 P1H Q2 • Jun 2023 P2H Q2 • Nov 2023 P1H Q4 • May 2024 P1HR Q2 • May 2024 P1H Q2 • Nov 2024 P2H Q5 • Jun 2025 P1HR Q4

 Your notes

? What is it?

In plain words

The probabilities in the table **add to 1**. Missing ones linked by a rule ("twice", " $4x$ ", "equal")? Write them with x and solve.

Then: "or" $\rightarrow$ **add**; "how many times" $\rightarrow$ $\times$ **trials**.

colour	blue	green	purple	red	yellow
P	0.12	0.20	0.38	$4x$	x
	$0.70 + 5x = 1 \Rightarrow x = 0.06$				

IT LOOKS LIKE

4x

x

"twice the probability"

How do I do it?

1 Known

Add the given probabilities.

2 Unknown

Write the missing ones with x .

3 Total 1

Solve for x .

4 Use it

Add for "or"; $\times$ trials for a count.

EVERYTHING ADDS TO 1 The most common variant.

V1 Probability Tables: Total 1

VARIANT 1 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“probability ... green is twice the probability ... red”

→ **Green = 2x, red = x**

“on blue or on green or on purple”

→ **Add them**

“spins the spinner 350 times ... estimate”

→ **$P \times 350$**

WORKED EXAMPLE PAST PAPER

June 2025 • Paper 1HR • Q4 • 5 marks

Blue 0.12, green 0.20, purple 0.38, red 4x, yellow x. (a) P(blue or green or purple). (b) Max spins 350 times. Estimate the number of reds.

STEP 1 $0.12 + 0.20 + 0.38 = 0.70$ ◀ (a) add

STEP 2 $0.70 + 5x = 1 \Rightarrow x = 0.06$ ◀ total 1

STEP 3 $P(\text{red}) = 4 \times 0.06 = 0.24$ ◀ red

STEP 4 $0.24 \times 350 = 84$ ◀ (b) expected

ANSWER (a) 0.70 (b) 84

✓ $0.70 + 0.24 + 0.06 = 1$ ✓

WATCH OUT

× $0.70 + 4x + x = 1$, then red = x

✓ red = $4x = 0.24$

Answer the colour asked, not just x.

EXAM TIP

Check your completed table adds to exactly 1.

Check yourself

calculator first • answers on page 13

1.1 May 2021 P1H

Red 0.20, blue 0.12, green 0.08; orange = $3 \times$ pink. P(orange)? Blue in 150 spins?

Answer:

1.2 May 2024 P1H

450 students: walk 0.20, bus x, car 2x, bicycle 0.26. How many travelled by car?

Answer:

1.3 Nov 2023 P1H

Blue 0.58, orange 2x, white x. Find x. Blue in 250 spins?

Answer:

V2

VARIANT 2

Expected & Relative Frequency

Expected number = probability $\times$ trials;
relative frequency = result $\div$ trials.

SEEN IN PAST PAPERS

3 / 32

PRACTICE QUESTIONS: 13


Your notes

Came up in: Oct/Nov 2020 P1H Q2 • Jan 2022 P1HR Q2 • May 2023 P1HR Q5

? What is it?

In plain words

If $P(\text{win}) = 0.65$, in 300 games **expect** about $0.65 \times 300 = 195$ wins.

From an experiment:

relative frequency = $\frac{\text{times}}{\text{trials}}$ — the more trials, the better the estimate.

IT LOOKS LIKE

0.65 $\times$ 300

faulty toy cars



$P = 0.7$: about 7 in every 10
in 300: about 210

How do I do it?

1 Probability

Of the event asked (lose = 1 – win).

2 Trials

How many times? (work it out if needed).

3 Multiply

$P \times$ trials.

4 Sense

An estimate — usually a whole number.

$P \times$ TRIALS Lose = 1 – win first.

V2 Expected & Relative Frequency

VARIANT 2 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Work out an estimate for the number of games that John wins” → 0.65×300

“estimate the number of games that Evie will lose” → $(1 - 0.74) \times 300$

“5 toy cars are made every 12 seconds” → **Trials = cars per day**

WORKED EXAMPLE PAST PAPER

Oct/Nov 2020 • Paper 1H • Q2 • 4 marks

Toy cars are made for 15 hours a day, 5 every 12 seconds. The probability a car is faulty is 0.002. Estimate the number of faulty cars made each day.

STEP 1 $15 \times 3600 = 54\,000$ s ◀ seconds in a day

STEP 2 $54\,000 \div 12 \times 5 = 22\,500$ ◀ cars a day

STEP 3 $0.002 \times 22\,500$ ◀ $P \times$ trials

STEP 4 $= 45$ ◀

ANSWER 45 faulty cars

✓ 2 in every 1000 ✓

WATCH OUT✗ 0.002×15 ✓ $0.002 \times 22\,500$

The trials are the cars, not the hours.

EXAM TIP

Expected frequency is an estimate — you don't need to round a whole-number answer.

Check yourself

calculator first • answers on page 13

2.1

Jan 2022 P1HR

$P(\text{John wins a game}) = 0.65$. He plays 300 games. Estimate his wins.

Answer:

2.2

May 2023 P1HR

$P(\text{Evie wins}) = 0.74$. She plays 300 games. Estimate how many she loses.

Answer:

2.3

Dr Eslam

$P(\text{win}) = 0.35$. Estimate the wins in 240 games.

Answer:

V3**VARIANT 3**

How Many in the Bag?

Probability = $\frac{\text{this colour}}{\text{total}}$: set up the fraction with the unknown and solve.

SEEN IN PAST PAPERS

3 / 32

PRACTICE QUESTIONS: 9



Your notes

Came up in: Nov 2021 P1H Q3 • Jun 2025 P2H Q11 • Nov 2025 P2H Q9

? What is it?

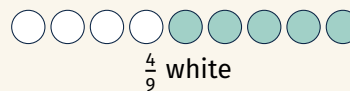
In plain words

$P(\text{white}) = \frac{4}{9}$ with 54 fish: white
 $= \frac{4}{9} \times 54 = 24$.

Unknown number in the fraction?

Write $\frac{\text{part}}{\text{total}} = P$ with the letter, and

cross-multiply. Adding items changes the **total** too.



IT LOOKS LIKE

$$\frac{24+k}{54+k} = \frac{1}{2}$$

How do I do it?

1 Fraction

$$\frac{\text{this kind}}{\text{all}} = P.$$

2 Unknown

Letter for what you don't know.

3 Solve

Cross-multiply.

4 Changes

Adding/removing changes top and bottom.

PART ÷ TOTAL = P Then cross-multiply.

V3 How Many in the Bag?

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"The probability that he takes a white fish is $\frac{4}{9}$ " → **White** = $\frac{4}{9} \times 54$

"He puts some more white fish into the tank" → $\frac{24+k}{54+k}$

Bags mixed together → **Count each colour, then ÷ the new total**

**WORKED EXAMPLE PAST PAPER**

November 2021 • Paper 1H • Q3 • 4 marks

54 fish; $P(\text{white}) = \frac{4}{9}$. (a) How many white? (b) Some white fish are added; now $P(\text{white}) = \frac{1}{2}$. How many were added?

STEP 1 $\frac{4}{9} \times 54 = 24$ ◀ (a)

STEP 2 $\frac{24+k}{54+k} = \frac{1}{2}$ ◀ k added to both

STEP 3 $48 + 2k = 54 + k$ ◀ cross-multiply

STEP 4 $k = 6$ ◀ (b)

ANSWER (a) 24 (b) 6✓ $\frac{30}{60} = \frac{1}{2}$ ✓**WATCH OUT**

✗ $\frac{24+k}{54} = \frac{1}{2}$
 ✓ $\frac{24+k}{54+k} = \frac{1}{2}$

The new fish are in the total too.

**EXAM TIP**

Check by putting your number back into the fraction.

**Check yourself**

calculator first • answers on page 13

3.1

Jun 2025 P2H

17 red, 28 blue, the rest orange; $P(\text{orange}) = \frac{4}{9}$.
How many orange?

Answer:

3.2

Nov 2025 P2H

Bag A: 120, $P(\text{red}) = 0.45$. Bag B: 80, $P(\text{red}) = 0.3$. All into bag C. $P(\text{red})$ from C?

Answer:

3.3

Dr Eslam

12 red counters and some blue; $P(\text{red}) = \frac{2}{5}$.
How many blue?

Answer:

V4**VARIANT 4** MOST COMMON IN THE EXAM

More Than One Pick

Multiply along each route; add the routes;
without replacement take one away from
top and bottom.

SEEN IN PAST PAPERS

13 / 32

PRACTICE QUESTIONS: 6



Your notes

Came up in: Jan 2020 P1H Q18 • Jan 2021 P2HR Q21 • May 2021 P2H Q23 • Jun 2022 P1HR Q24 • Jun 2022 P1H Q15 • Jun 2023 P2HR Q15 • Jun 2023 P2HR Q19 • Nov 2023 P1H Q20 • May 2024 P1HR Q21 • Jun 2024 P2HR Q14 • Jun 2025 P1HR Q19 • Jun 2025 P1H Q21

? What is it?

In plain words

One pick after another: **multiply**.

Different ways to get the result (red then blue, blue then red): **add** them.

Without replacement: the second fraction has one fewer on the bottom (and on top, if the same colour).

$$P(RR) = \frac{5}{8} \times \frac{4}{7}$$

$$P(RB) = \frac{5}{8} \times \frac{3}{7}$$

$$P(BR) = \frac{3}{8} \times \frac{5}{7}$$

IT LOOKS LIKE

3 sweets taken

without replacing

ONE LESS ON THE BOTTOM

How do I do it?

1 Routes

List every order that gives the result.

2 Each route

Multiply the fractions (adjust for no replacement).

3 Add

All the routes.

4 Algebra?

Set equal to the given probability; solve.

*** ALONG, + BETWEEN** The most common late question.

V4 More Than One Pick

VARIANT 4 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“takes at random ... and does not replace”

→ **Bottom goes down by 1**

“the sum of the numbers ... is less than 5”

→ **List the pairs; both orders**

“The probability ... is $\frac{9}{17}$. Find n ”

→ **Expression in n = given; quadratic****WORKED EXAMPLE PAST PAPER**

June 2025 • Paper 1HR • Q19 • 3 marks

Counters numbered 1, 1, 2, 3, 3, 3, 5, 6, 6. Two are taken without replacement. Find the probability that the sum is less than 5.

STEP 1 (1, 1), (1, 2), (2, 1), (1, 3), (3, 1) ◀ sums under 5

STEP 2 $\frac{2}{9} \cdot \frac{1}{8} + 2 \cdot \frac{2}{9} \cdot \frac{1}{8}$ ◀ (1, 1); (1, 2) both orders

STEP 3 $+ 2 \cdot \frac{2}{9} \cdot \frac{3}{8} = \frac{2+4+12}{72}$ ◀ (1, 3) both orders

STEP 4 $= \frac{18}{72} = \frac{1}{4}$ ◀ simplify

ANSWER $\frac{1}{4}$

✓ list of 72 ordered pairs gives 18 ✓

**WATCH OUT**

✗ only one order of (1, 2)

✓ both (1, 2) and (2, 1)

Different orders are different routes.

**EXAM TIP**

Write each route on its own line, then add.

**Check yourself**

calculator first • answers on page 13

4.1

Jan 2020 P1H

16 sweets: 4 blackcurrant, 5 lemon, 7 orange. Three are taken. P(all 5 lemons still in the bowl)?

Answer:

4.2

Jun 2023 P2HR

A biased coin lands heads three times in three spins with probability $\frac{27}{64}$. P(three tails)?

Answer:

4.3

Jun 2024 P2HR

P(heads) = 0.8; P(6 and heads) = 0.24. P(6 and tails)?

Answer:

CHECK YOURSELF

Answers



Every answer below was checked in Python (check_answers.py).

1.1 orange 0.45; blue
18

1.2 162

1.3 $x = 0.14$; 145

2.1 195

2.2 78

2.3 84

3.1 36

3.2 0.39

3.3 18

4.1 $\frac{33}{112}$

4.2 $\frac{1}{64}$

4.3 0.06

THE WHOLE TOPIC ON ONE PAGE

Probability cheat sheet



Your notes
V1 Tables 13/32

$$0.70 + 5x = 1$$

Total 1; then $\times$ trials.**V2 Expected 3/32**

$$0.65 \times 300$$

 $P \times$ trials.**V3 In the bag 3/32**

$$\frac{24+k}{54+k} = \frac{1}{2}$$

Part $\div$ total; cross-multiply.**V4 Two picks 13/32**

$$\frac{5}{8} \times \frac{4}{7}$$

 $\times$ along, + between.**TOP 3 LOST MARKS**

1. giving x instead of $4x \times$
2. forgetting one order $\times$
3. no change without replacement $\times$

BOTTOM LINE Total 1 · $P \times$ trials · part $\div$ total · multiply along, add between.